

GENERAL SPECIFICATION G3.20

Concrete

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G3.20 CONCRETE

G3.20.1 Scope

- (a) This Part covers the requirements for the supply of materials for concrete; design of mixes; quality control; mixing, transporting and placing; and curing of concrete; requirements for joints in concrete and surface protection of concrete by membranes against groundwater and the salts which may be dissolved thereon.
- (b) Further requirements specific to the construction of water retaining structures and the building of pipework and other items are given in the Particular Specification.

G3.20.2 Reference Standards

- (a) Unless otherwise specified, materials for concrete, concrete products and testing procedures shall comply with the following standards:

Standard	Subject
ASTM D 512	Tests for chloride ion in water and waste water
ASTM D 516	Tests for sulphate ion in water and waste water
ASTM D 545	Testing preformed expansion joint fillers for concrete construction (non-extruding and resilient types)
SS EN 1990	Basis of structural design
SS EN 1991-1-5	Actions on structures: Thermal actions
SS EN 1991-1-6	Actions on structures: Actions during execution
SS EN 1992	Design of concrete structures
SS EN 1997	Geotechnical design
SS 197-1	Cement: Composition, specification, and conformity criteria for common cements
SS EN 206-1	Concrete: Specification, performance, production and conformity
SS 544	Complementary Singapore Standard to SS EN 206-1
BS EN 12350	Testing fresh concrete
BS EN 12390	Testing hardened concrete
BS EN 10080	Steel for reinforcement of concrete
BS 4449	Steel for the reinforcement of concrete. Weldable reinforcing steel. Bar, coil and decoiled product
BS 4483	Steel fabric for the reinforcement of concrete
SS 560	Steel for reinforcement of concrete – Weldable reinforcing steel – Bar, coil and decoiled product
SS 561	Steel fabric for the reinforcement of concrete
BS 5896	Specification for high tensile steel wire and strand for the prestressing of concrete
BS 4486	Specification for hot rolled and processed high tensile alloy steel bars for the prestressing of concrete

Standard	Subject
BS EN 10138	Pre-stressing steels
BS EN ISO 17660	Welding – Welding of reinforcing steel
BS EN 13670	Execution of Concrete Structures
BS EN 12504	Testing concrete in structures
SS EN 13791	Assessment of in-situ compressive strength in structures and precast concrete components
BS EN ISO 15630	Steel for the reinforcement and pre-stressing of concrete: test methods
BS EN 15743	Supersulphated Cement
BS 6213	Guide to selection of construction sealants
BS 6949	Bitumen based coatings for cold application excluding use in contact with potable water
SS EN 12620	Aggregates for concrete
BS 812	Testing aggregates
BS 8102	Protection of structures against water from the ground
BS 8666	Scheduling, dimensioning, bending, and cutting of steel reinforcement for concrete

G3.20.3 Submissions by the Contractor

- (a) Introduction
 - (i) Submissions by the Contractor in relation to concrete work are described in the relevant clauses of the Specification. These submissions are summarised hereunder.
- (b) Certificates
 - (i) Manufacturers' and suppliers' certificates of compliance with relevant standards in respect of the following materials:
 - 1) Cement;
 - 2) Aggregates;
 - 3) Admixtures;
 - 4) Jointing materials; and
 - 5) Ready mixed concrete.
 - (ii) Calibration certificates furnished by an approved person for:
 - 1) Weighing devices;
 - 2) Dispensing devices; and
 - 3) Batching plant.

- (iii) The results certified by an approved person for tests carried out subsequently on:
 - 1) Aggregate;
 - 2) Moisture content of aggregate;
 - 3) Fresh concrete;
 - 4) Concrete test tubes; and
 - 5) Hardened concrete.
- (iv) Samples
 - 1) Samples of all the foregoing materials except concrete and water shall be submitted, and those approved shall be kept in suitable containers, properly labelled and stored on Site for reference.
- (v) Concrete Mixes
 - 1) Details of designed mixes for all grades of concrete required for the Works, together with proposed methods of transportation, handling, placing and compaction.
- (vi) Other Submissions
 - 1) Proposals for the following, where relevant to the Works:
 - Details of onsite concrete production plant;
 - Details of designed mixes;
 - Layout of construction jointing and lifts;
 - Special measures for dealing with particular circumstances (e.g. concreting in hot weather);
 - Methods for precast work;
 - Methods for prestressed work; and
 - Methods of curing.
- (c) Records of Concreting
 - (i) The Contractor shall instigate a concrete "pour card" system which shall be applied to every concrete pour, whether reinforced concrete or mass concrete. Prior to every pour, the Contractor shall submit to the S.O. the relevant pour card which shall have been signed off by a member of the Contractor's engineering staff indicating that the various components of the work including formwork, falsework, reinforcement, other necessary preparations for concreting, etc., have been checked by a responsible member of the Contractor's staff. The format of the pour card shall be to the approval of the S.O..
 - (ii) Daily returns in respect of all concrete placed during the previous day.

- (iii) The returns shall detail:
- 1) In respect of each grade of concrete:
 - The number of batches mixed;
 - The number of batches and the total volume of concrete placed;
 - The number of batches wasted or rejected; and
 - The weight of cement used.
 - 2) In respect of each location in the Works:
 - The position of the pour (e.g. bay or lift reference number);
 - The grade of the concrete placed; and
 - The total volume of concrete placed and the number of batches used.
- (iv) In addition, the Contractor shall maintain an accurate and up to date record showing dates, times, weather and temperature conditions when each part of the Works was concreted. The record shall be available for inspection by the S.O. at all times.
- (v) Results of all tests on concrete shall be recorded and identified with the parts of the Works to which they relate.

G3.20.4 Materials

- (a) Cement
- (i) Cement used in the Works shall be ordinary Portland cement unless otherwise specified or agreed with the S.O..
 - (ii) The Contractor shall provide from each consignment of cement delivered to the Site such samples as the S.O. may require for testing. Any cement which is, in the opinion of the S.O., lumpy or partially set shall be rejected and the Contractor shall promptly remove such cement from the Site.
 - (iii) Cement which has been stored on the Site for more than forty days and cement which in the opinion of the S.O. is of doubtful quality shall not be used in the Works unless it is retested and the test results show that it complies with the relevant Standard.
- (b) Storage of cement
- (i) Immediately upon arrival at the Site, cement shall be stored in silos designed for the purpose or in dry weather-tight and properly ventilated structures with floors raised 500 mm above ground level with adequate provision to prevent absorption of moisture. All storage facilities shall be subject to approval by the S.O. and shall be such as to permit easy access for inspection and identification. Each consignment of cement shall be kept separately and the Contractor shall use the consignments in the order in which they are received.

- (ii) Cement of different types and from different sources shall be kept in clearly marked separate storage facilities. Cement delivered to the Site in drums or bags provided by the supplier or manufacturer shall be stored in the unopened drums or bags until used in the Works. Any cement in drums or bags which have been opened on the Site shall be used immediately or shall be removed from the Site.
- (c) **Aggregates**
 - (i) Aggregates for concrete shall be obtained from an approved source, shall conform to the grading and other requirements of the relevant standards, and shall be washed clean.
 - (ii) Water absorption of aggregates, when tested in accordance with the relevant standard, shall not exceed 3%.
 - (iii) **Mine Waste**
 - 1) Aggregates for concrete shall not be obtained from the wastes of metalliferous mining.
 - (iv) **Alkali-silica Reaction**
 - 1) The fine and coarse aggregates shall each consist of at least 95% of one or more of the rock types tabulated below. Sources of aggregates shall not contain opaline silica or quantities of flint, chert or chalcedony that could cause damage from alkali-silica reaction.

Air cooled blastfurnace slag	Expanded clay/ shale/slate	Microgranite
Andesite	Feldspar	Quartz
Basalt	Gabbro	Schist
Diorite	Gneiss	Sintered PFA
Dolerite	Granite	Slate
	Limestone	Syenite
 - 2) If the Contractor wishes to use aggregates other than those listed, he shall submit evidence to the satisfaction of the S.O. from the previous performance of the coarse and fine aggregates which he proposes to use that, when used with the proposed cementitious constituents of the concrete mix in the proportions proposed there will be no cracking or expansion due to alkali-silica reaction during the life of the structure.
 - 3) If this requirement cannot be met the Contractor shall adopt constituents for his concrete such that either
 - The cementitious material shall have a reactive alkali content not exceeding a maximum value of 0.6% by mass when defined and tested in accordance with the method prescribed, or
 - The total mass of reactive alkali in the concrete mix shall not exceed 3 kg per m³ of concrete when defined, tested and calculated in accordance with the method prescribed.

- 4) The Contractor shall notify the S.O. of his proposals for complying with this requirement at the time of tendering.
 - 5) The methods prescribed for complying with 3(a) and 3(b) above shall be those set out in:
 - Alkali-Silica Reaction: Concrete Society Technical Report No 30, October 1987.

For requirement

(a) Clauses 16 and 17; and

(b) Clauses 16 and 18.
 - 6) Soluble Chlorides
 - Chlorides content, expressed as chloride ion, shall be such that the concrete mix as a whole complies with the limit of total chlorides stated in this specification.
 - 7) Soluble Sulphates
 - Water-soluble sulphate contents of the aggregates shall be such that the concrete mix as a whole complies with the limit of total water soluble sulphates stated in this specification. If necessary the tests described in BS 812: Part 118 for acid soluble sulphates shall be used to ensure compliance with this requirement.
 - 8) Coarse and Fine Low-shrinkage Aggregates
 - The coarse aggregates shall be capable of producing concrete having a drying shrinkage of not more than 0.065%, when tested in accordance with the method given in BS 812: Part 120;
 - The Contractor shall produce test sheets from the supplier certifying compliance with the stipulated requirement. Where such test sheets are not available, tests shall be carried out by the Contractor to establish the suitability of the proposed source of supply; and
 - Where a source of supply produces coarse aggregates which do not satisfy the stipulated shrinkage requirement, such a source shall not be used for the supply of fine aggregates.
- (d) Storage of Aggregates
- (i) The Contractor shall provide means of storing the aggregates at each point where concrete is made such that (i) each nominal size of coarse aggregate and the fine aggregate shall be kept separated at all times, (ii) contamination of the aggregates by the ground or other foreign matter shall be prevented at all times, and (iii) each heap of aggregate shall be capable of draining freely;
 - (ii) The Contractor shall ensure that graded coarse aggregates are tipped, stored and removed from store in a manner that does not cause segregation; and

- (iii) Wet fine aggregate shall not be used until in the opinion of the S.O. it has drained to a constant and uniform moisture content, unless the Contractor measures the moisture content of the fine aggregate continuously and adjusts the amounts of fine aggregate and added water in each batch of concrete mixed to allow for the water contained in the fine aggregate. If necessary to meet the requirements of this clause, the Contractor shall protect the heaps of fine aggregate against inclement weather.
- (e) Water
 - (i) Water for washing aggregates, for mixing concrete and for curing shall be clean and free from harmful matter; the concentration of sulphates and chlorides shall be such that the concrete mix as a whole complies with the specified limits of salts content.
- (f) Admixtures
 - (i) Admixtures shall mean material added to the concrete materials during mixing for the purpose of altering the properties of the concrete mix.
 - (ii) Admixtures containing calcium chloride shall not be used.
 - (iii) The chloride ion content of admixtures shall not exceed 2% by weight of the admixture or 0.03% by weight of the cement when used in concrete containing prestressing tendons, reinforcement or embedded metal.
 - (iv) Admixtures shall be used only if the S.O. has given his prior approval in writing, and with due regard to the manufacturers' instructions. Both the amount added and the method of use shall be to the approval of the S.O. who shall also be provided in good time with the following information:
 - 1) The typical amount added and any detrimental effects due to adding greater or smaller amounts;
 - 2) The chemical name(s) of the main active ingredient(s) in the admixture; and
 - 3) Whether or not the admixture leads to the entrainment of air when used at the proposed rates of dosage.
 - (v) When more than one admixture is used in a concrete mix the compatibility of the various admixtures shall be established to the satisfaction of the S.O..
- (g) Jointing Materials
 - (i) Jointing materials shall include water stops, fillers, paints, caulking compounds, sealants, adhesives and other such materials required for the making of joints in concrete
 - (ii) Jointing materials shall be obtained from approved manufacturers. They shall be handled and stored in accordance with the manufacturer's recommendations; and

- (iii) Where there is no appropriate standard to which reference may be made in respect of any proprietary jointing material, the Contractor shall prove by demonstration, tests or otherwise the suitability of the material under Site conditions. In other instances, the Contractor shall supply manufacturers' test sheets as evidence of compliance with relevant quality standards. Only such materials as have been approved by the S.O. shall be used in the Works.
- (h) Epoxy Concrete Bonding Agent
 - (i) Where specified or indicated on the Drawings, an epoxy concrete bonding agent shall be used for bonding new cementitious materials to existing cementitious surfaces. The bonding agent shall comprise a two part solvent free epoxy resin formulation approved by the S.O..

G3.20.5 Workmanship

- (a) Free Water/Cement Ratio
 - (i) In designing and establishing approved mixes of concrete for any part of the Works the Contractor shall keep within any limitations on free water/cement ratios which may be expressly stated in the Specification, or shown on the Drawings as applying to concrete for particular parts of the Works or (where not so stated or shown) as given in the Table of concrete grades.
- (b) Limits of Salts Content of Mix
 - (i) No concrete shall contain more than the following total quantities of the substances expressed as percentages by weight of cement:
 - 1) The total chlorides expressed as chloride ions calculated whenever possible from the mix proportions and the measured chloride contents of each of the constituents: 0.4%;
 - 2) For sulphate –resisting or super sulphated cements the total chlorides expressed as chloride ions calculated as above: 0.2%;
 - 3) For mixes used for prestressed concrete or heat-cured concrete containing embedded metal (all types of cement), the total chlorides expressed as chloride ions calculated as above: 0.1%; and
 - 4) For all mixes except where supersulphated cement is used, total water-soluble sulphates: 4.0% (as SO₃ ions).
 - (ii) Tests shall be carried out in accordance with the following standards:
 - 1) Aggregates
 - Chlorides BS 812 Part 117; and
 - Sulphates BS 812 Part 118.
 - 2) Mixing water
 - Chlorides ASTM D512; and
 - Sulphates ASTM D516.
 - 3) Calcium chloride or chloride-based admixtures shall not be added to mixes intended to contain reinforcement, pre-stressing tendons or other embedded metal.

- 4) Where it is required to verify the salt content of hardened concrete, tests shall be carried out in accordance with BS EN 12390.
- (c) Workability
 - (i) The workability of the concrete mixes shall be defined by their compacting factors determined by the method described in BS EN 12350-4.
 - (ii) The workability of each concrete mix shall be such that satisfactory compaction can be obtained when the concrete is placed and vibrated in the work and that there is no tendency to segregate when it is handled, transported and compacted by the methods which the Contractor proposes to use in the Works.
 - (iii) When a mix has been verified as complying with these requirements its workability shall be measured and, within the limits given below, confirmed or amended as considered necessary by the S.O..
 - (iv) For reinforced concrete and prestressed concrete, the compacting factor shall lie between 0.85 and 0.92. For un-reinforced concrete, the compacting factor shall lie between 0.75 and 0.80. For concrete which is to be pumped, the compacting factor may be increased to 0.95.
- (d) Design of Concrete Mixes
 - (i) Each mix design shall be such that:
 - 1) The aggregate shall comprise both fine aggregate and coarse aggregate. The maximum size of coarse aggregate shall be either 20 mm or 40 mm as directed by the S.O.. A separate mix shall be designed for each maximum aggregate size for any grade of concrete;
 - 2) The maximum free water/cement ratio shall be the maximum water/cement ratio when the aggregate is saturated but surface dry; and
 - 3) The mixes shall be designed to produce a target mean concrete cube strength at 28 days after manufacture greater than the characteristic strength at 28 days by a margin of at least 6 N/mm² for grade 15 and a margin of at least 8 N/mm² for grades 20, 25, 30, 35, 40 and 45.
 - (ii) For any concrete containing admixtures, the strengths shall be not less than those specified in the Table of concrete grades but the mixes shall be separately designed to take account of the effects of the admixtures, and shall have separate trial mixes made and tested.

(e) Trial Mixes

(i) Laboratory trials:

- 1) As soon as the S.O. has agreed the trial mix proportions for each grade of concrete, the Contractor shall produce in the concrete-testing laboratory on Site two batches from a trial mix for each grade (except grade 10) using cement and surface dry aggregates known to be typical of the proposed source of supply. Each batch shall contain the correct amount of cement and have a free water/cement ratio at or below the maximum value given in the table of concrete grades, and the workability of each batch shall be determined. Provided that the compacting factors so obtained fall within the specified limits and the requirements regarding cement content and free water cement ratio have been met, the Contractor may proceed to carry out Site trials.
- 2) As an alternative to producing trial mixes in a laboratory as described above, the Contractor may submit recent existing laboratory test reports from elsewhere. Such reports shall be acceptable only if the S.O. is satisfied with the laboratory's test procedures and that the materials used were typical of the proposed source of supply.

(ii) Site trials:

- 1) Trial mixes shall be made at the site of the proposed concreting under full scale production conditions. The proportions of materials used, including the mixing water, shall agree with the mix proportions determined by the laboratory trial mixes. Three batches of concrete for each grade shall be produced, using the same plant and mixing time proposed by the Contractor for use in the Works.
- 2) For each grade of concrete the following shall be carried out:
 - The workability of each of the three batches shall be determined immediately after mixing by the compacting factor method described in BS EN 12350; and
 - Three 150 mm test cubes from each of the three batches shall be made by the Contractor in the presence of the S.O.. The cubes shall be made, cured, stored and tested at 28 days after manufacture in accordance with the method described in BS EN 12390.
- 3) A site trial mix for a particular grade of concrete shall be considered satisfactory provided that:
 - The compacting factors as under (a) above fall within the specified limits; and
 - The average value of the compressive strength of the nine cubes taken from the trial mix is greater than the target mean strength used in the mix design.
- 4) If the requirements under (i) and (ii) above are not met for any mix the Contractor shall re-design that mix and he shall make further site trials as above.

- 5) Where re-design of any concrete mix becomes necessary, the making and testing of the trial mix shall be repeated until the trial mix satisfies both requirements (i) and (ii) above. Final approval will not be given to any trial mix which does not comply with the specified requirements appropriate to that grade of concrete.
- (f) Mixes Not Approved
- (i) Approval of a mix may be withheld or withdrawn if (inter alia):
 - 1) the grading of the aggregate changes such that the fraction of aggregate retained on any sieve differs from the corresponding fraction of aggregate in the approved mix by more than 2% of the total quantity of fine and coarse aggregate; or
 - 2) the source of supply of aggregate or cement is changed.
 - If approval of a mix for any grade of concrete is withdrawn for any reason the Contractor shall carry out such further trials as are necessary to achieve a satisfactory mix for that grade of concrete.
- (g) Batching of Materials
- (i) Cement used in the production of concrete shall be measured by weight either with an approved weighing machine or by making the size of each batch of concrete such as to require an integral number of complete bags or drums of cement.
 - (ii) For concrete of grades greater than C12/15 the fine aggregate and the several nominal sizes of coarse aggregate shall be measured singly or cumulatively by weight using weigh-batching machines.
 - (iii) For concrete of grades less than C12/15 the fine and coarse aggregate shall be measured separately either by weight using weigh-batching machines or by volume in gauge boxes.
 - (iv) Weigh-batching machines shall provide facilities for the accurate control and measurement of the aggregates either singly or cumulatively and shall be capable of immediate adjustment by semi-skilled operators in order to permit variations to be made to the mix. All weigh dials shall be easily visible from the place at which filling and emptying of the hoppers is controlled.
 - (v) Every concrete mixing machine shall be fitted with a device to measure added water by weight or volume and shall be so constructed that the water inlet and outlet valves are interlocked so that neither one of them can be opened unless the other is fully closed. The water-measuring device shall be provided with an overflow with a cross-sectional area at least four times that of the inlet pipe and with its discharge point clear of the mixing plant. The entire water system shall be maintained free of leaks at all times and the measuring device shall be fitted with a drain pipe which allows the full quantity of water being measured to be drained off for checking the measurement. The outlet arrangement of the measuring device shall be such that between five and ten percent of the water enters the mixer before the other materials and a further five to ten percent of the water enters the mixer after the other materials. The remainder of the water shall be added at a uniform rate with the other materials. The water measuring device shall be readily adjustable so that the quantity of water added to the mixer can, if necessary, be varied for each batch.

- (vi) Gauge boxes for use in volume batching shall be soundly constructed of timber or steel. Each box shall be sized to contain exactly the volume of aggregate required for one batch of any particular mix. In sizing up gauge boxes for fine aggregates, the Contractor shall make an allowance for bulking due to moisture contained in the aggregate stockpiles on Site. Each box shall be suitably identified by reference to the aggregate and the mix for which it has been made.
- (vii) Any admixtures which may be used shall be measured separately in calibrated dispensers.
- (viii) All mixing and batching plants shall be maintained free of set concrete or cement and shall be clean before commencing mixing. The accuracy of calibration of any weighing plant, water measuring device and admixture dispenser shall be checked before carrying out trial mixes, before the first mixing of concrete for inclusion in the Works, after each service or adjustment to the mixing plant, and in any case at least once per week.
- (h) Mixing Concrete
 - (i) Concrete shall be mixed in batches in plant capable of combining the aggregates, cement and water (including admixtures if any) into a mixture uniform in colour and consistency, and of discharging the mixture without segregation.
 - (ii) On commencing work with a clean mixer the first batch shall contain only half the normal quantity of coarse aggregate to compensate for the adhesion of the other materials to the drum.
 - (iii) The moisture contents of the aggregates shall be determined before the commencement of each day's concreting and at such intervals during each day as may be necessary. The Contractor shall make due allowance for water contained in the aggregates when determining the quantity of water to be added to each mix, and shall adjust the amount of water added to each mix to maintain the approved free water/cement ratio of the mixed concrete.
- (i) Ready-Mixed Concrete
 - (i) Any concrete which is not prepared under the direct control of the Contractor and by plant situated on or reasonably adjacent to the Site shall be classed as ready-mixed concrete.
 - (ii) Ready-mixed concrete shall not be used in any part of the Works without the written approval of the S.O., which approval may be withdrawn at any time.
 - (iii) The Contractor shall satisfy the S.O. that ready-mixed concrete complies with the Specification for concrete, and that the manufacturing and delivery resources of the proposed supplier adequate to ensure proper and timely completion of each concreting operation. The proposed supply depot shall be open to inspection by the S.O. at all times.
 - (iv) The specified requirements as to the sampling, trial-mixing, testing and quality of concrete of various grades shall apply equally to ready-mixed concrete which shall furthermore be made and delivered in accordance with the relevant standards.

- (v) The Contractor shall provide every additional facility which the S.O. may require for the supervision and inspection of the batching, mixing and transporting of ready-mixed concrete.
- (j) Preparing for Concreting
 - (i) Before placing concrete the Contractor shall remove from the surface of the foundations or previously placed concrete all oil, loose fragments of rock, earth, mud, timber or other debris and any standing water.
 - (ii) Where specified, and elsewhere as ordered by the S.O., excavated surfaces on which concrete is to be placed shall be covered with either blinding concrete not less than 75 mm thick, or waterproof building paper, or polythene sheeting 0.1 mm in thickness immediately after completion of the final trimming of the excavation.
- (k) Transporting Concrete
 - (i) Concrete shall be conveyed from the mixer to its place in the Works as rapidly as possible by methods which will prevent segregation or drying out and ensure that the concrete is of the required workability at the time of placing. If segregation has nevertheless occurred in any instance the materials shall be remixed or rejected.
- (l) Placing Concrete
 - (i) Before placing concrete in any part of the Works the Contractor shall satisfy himself by inspection that the part is in all respects ready for the reception of concrete. He shall also notify the S.O. providing at least 4 working hours' notice so as to enable him to inspect the location if so required.
 - (ii) Concrete shall be placed and compacted before initial set has occurred and, in any event, not later than forty-five minutes from the time of mixing.
 - (iii) Generally concrete shall be carefully placed in horizontal layers which shall be kept at an even height throughout the work. Concrete shall not be allowed to slide or flow down sloping surfaces into its final position but shall be placed directly in its final position from skip, truck, barrow, down pipe or other placing machine or device or, if this is impossible, it shall be shovelled into position with care being taken to avoid segregation. Concrete placed in horizontal slabs from barrows or other tipping vehicles shall be tipped into the face of the previously placed concrete.
 - (iv) For large, deep slabs, the Contractor shall plan the pouring sequence to limit the size of the working face. This procedure shall be to the approval of the S.O..
 - (v) Concrete dropped into place in the work shall be dropped vertically. It shall not strike the formwork between the point of its discharge and its final place in the work. Concrete shall not be dropped freely through a height greater than 1½ m. Chutes and conveyor belts shall be so designed that there is no segregation or loss of mortar and shall be provided with a vertical pipe, or other device, which ensures that concrete is discharged vertically into place.
 - (vi) Where a lift of concrete is built up in layers, each layer shall be properly merged into the preceding layer before initial set takes place. If necessary the area of the pour will be restricted to ensure that this is achieved.

- (vii) When pneumatic placers are used, if the end of the placer pipe is not equipped with an energy absorbing device it shall be kept as close to the work as practicable. Mortar or water used at the beginning or end of a run shall be discharged outside the formwork.
 - (viii) When pumps are used the end of the supply pipe shall be kept immersed in the concrete during placing to assist compaction. Mortar or water used at the beginning or end of a run shall be discharged outside the formwork.
 - (ix) Where concrete abuts against earth or other materials liable to become loose or to slip, the Contractor shall take steps to prevent any such loose material falling on to the surface of the concrete. Subject to the approval of the S.O. such steps may include leaving timbering in place or cutting away and removing timbering in small lengths or depths at a time.
- (m) Concreting in Hot Weather
- (i) Hot weather tends to impair the quality of fresh or hardened concrete or otherwise resulting in abnormal properties. In hot weather, the Contractor shall ensure that the temperature of the concrete as placed shall not exceed 32 degrees Celsius and that the maximum internal temperature attained during setting does not exceed 70 degrees Celsius.
 - (ii) To that end the Contractor shall provide sun shades over stockpiles of aggregates, cement silos, mixing water tanks and pipelines, and in addition shall carry out one or more of the following procedures which shall be submitted to the S.O. for approval.
 - (iii) Cool the mixing water and/or replace part of the water by chipped ice. The ice shall be completely melted by the time mixing is completed.
 - (iv) Spray clean cool water over the aggregate stockpiles. The Contractor shall carry out regular tests on the aggregates to ensure that concentrations of sulphates or chlorides do not rise to unacceptable levels, and to ensure that moisture content determinations allow for such spraying.
 - (v) Shade or wet the outside of the formwork.
 - (vi) Apply a fine moisture (fog) spray of clean cool water in order to cool and moisten the surrounding air and the sub-surface, to cool the formwork and reinforcement, and to lessen rapid evaporation from unformed concrete surfaces.
 - (vii) Pour concrete at night.
 - (viii) Change the mix to include retarding/plasticising admixtures, cement with a lower rate of hydration, or cement replacements.
 - (ix) The Contractor shall provide the S.O. with details of the precautions he proposes to take to protect the concrete from the effects of high temperatures. No concreting in hot weather shall be put in hand until the proposed measures have been approved by the S.O..

- (n) Concreting of Thick Sections
- (i) Thick concrete section shall be defined as any structural element with its thickness exceeding 500mm. The Contractor shall take all necessary precautions and measures according to proven practices in concrete technology to prevent any form of cracking due to temperature and shrinkage effects in casting of thick concrete sections. These shall be proposed by the Contractor and may include one or more of the following:
 - (ii) Ground granulated blast furnace slag, pfa or silica fume as a cement substitute to reduce heat of hydration and to increase durability and water tightness of concrete.
 - (iii) Other admixtures to increase workability, improve durability and reduce heat of hydration if necessary. The admixtures to be used in concrete shall not contain chlorides.
 - (iv) Use of flaked ice or liquid nitrogen as an additive to the concrete mix.
 - (v) A coolant system to contain the thermal difference and minimise thermal cracking of concrete where necessary.
 - (vi) Controlled insulation of the concrete during curing:
 - 1) Delay in striking of formwork.
 - 2) The temperature differential between the warmer interior portion and the cooler surface portion of the thick element shall be limited to prevent early thermal cracking due to heat of hydration. The allowable differential shall take into consideration the boundary restraint of the structural element. The peak temperature gradient from the core to any external surface of a concrete pour during the first two weeks after casting shall not exceed 20°C, unless the Contractor can demonstrate to the satisfaction of the S.O. that the proposed alternative will not have any deleterious effects on the concrete element, such as unacceptable internal or external cracking, or reduced long term durability. The maximum temperature within the element shall not exceed 70°C.
 - 3) The Contractor shall propose a method of placing, curing, temperature monitoring and any other measures for acceptance by the Engineer. The Contractor shall demonstrate that his proposal shall be adequate to prevent crack arising from heat of hydration and drying shrinkage. The design concrete mix submitted for acceptance by the Contractor shall take the above requirements into account.
 - 4) In the event that GGBS is being added as cement substitute at not less than 60% by mass to reduce the heat of hydration, a maximum temperature within the element not exceeding 75°C may be considered for acceptance by the Engineer.
 - 5) The Contractor shall carry out all necessary tests to demonstrate to the acceptance of the Engineer, that his proposed mix design together with his proposals to limit temperature differentials and to prevent early thermal cracking are satisfactory.

- 6) The Contractor shall provide and install thermocouples with electronic data recording equipment and any other necessary equipment and instruments to measure and monitor the concrete temperatures. The Contractor shall propose the location and the number of points, for thermocouples for the acceptance of the Engineer.
 - 7) Notwithstanding the above measures and precautions the Contractor shall where practical seal, repair, and rectify any cracks in any concrete elements with epoxy resins or other accepted materials at his own expense.
- (vii) Concrete Temperature and Site Monitoring
- 1) Methods of assessment of the in situ temperature rise and monitoring of pour temperatures shall be subject to approval of the S.O.. A thermal block shall be cast and monitored for all concrete grades C32/40 and above
 - 2) Thermal Block:
 - The thermal test blocks shall comprise 1.0 m x 1.0 m x 1.0 m cubes of unreinforced concrete. The blocks shall be cast within plywood formwork lined on all surfaces with 0.15 m thick polystyrene, to allow free expansion and contraction of the concrete and to minimise temperature losses.
 - A set of two thermocouples shall be placed at the following locations: the geometric centre, top corner, side mid position and top mid position. The temperature data shall be recorded by data loggers at half-hour intervals as a minimum until the temperature falls to ambient.
- (o) Precautions for Thick Sections
- (i) For concrete sections thicker than 500mm, additional precautions shall be taken as defined in concreting of thick sections under clause G.3.20.5 (n) of general specification. The effectiveness of these precautions shall be assessed in every pour by monitoring the concrete temperature using thermocouples as follows:
 - 1) Three locations shall be monitored for each pour at positions to be determined by the Engineer;
 - 2) At each location, the following temperatures shall be monitored:
 - Centre of the concrete section; and
 - 50 mm from an exposed surface;(i.e. walls: near face and far face: base slab: top)
 - 3) Monitoring shall commence from the time concrete is first discharged and may cease when the core temperature has fallen to 55°C; and

- 4) Monitoring may be discontinued in subsequent pours of similar structural elements if the Contractor can demonstrate that the requirements of clause G3.20.5 (n) are being met and that no thermal cracking of the concrete is occurring. Monitoring will be required again whenever in the opinion of the Engineer the nature of the concrete element being cast is changed or the precautionary measures initially adopted are modified.
- (p) Concrete Placed in Water
 - (i) Where any concrete is to be placed in water, the Contractor shall submit detailed proposals to the S.O. and shall obtain his approval before commencing the work.
 - (ii) The quantity of cement in any concrete placed in water shall if necessary be increased so that the free water/cement ratio of the mix is not more than 0.47.
 - (iii) Concrete shall not be placed in running water or be allowed to fall through water.
 - (iv) Concrete shall be placed in water only by means of a bottom-opening watertight box or a tremie of a type approved by the S.O.. Bottom-opening boxes shall not be opened until they are resting on the work, and the lower ends of tremies shall always be kept below the surface of freshly placed concrete.
- (q) Concreting during heavy downpour
 - (i) Contractor shall protect the freshly laid concrete from rain by using suitable cover or other means so that the integrity of concrete is preserved.
- (r) Compaction
 - (i) Except as otherwise approved by the S.O., concrete placed in situ shall be compacted with power driven internal type vibrators supplemented by hand spading and tamping, and shall be thoroughly worked around the reinforcement, tendons, duct formers and embedded items, and into the corners of the formwork, to form a solid mass free of voids. The vibrators shall always be adequate in number, amplitude and power to compact the concrete properly and quickly throughout the whole of the volume being compacted. Spare vibrators shall be readily on hand in case of breakdown.
 - (ii) Vibrators shall be inserted into the uncompacted concrete vertically and at regular intervals. Where the uncompacted concrete is in a layer above freshly compacted concrete the vibrator shall be allowed to penetrate vertically for about 100 mm into the previous layer. In no circumstances shall vibrators be allowed to come into contact with the reinforcement or formwork nor shall they be withdrawn quickly from the mass of concrete but shall be drawn back slowly so as to leave no voids. Internal type vibrators shall not be placed in the concrete in a random or haphazard manner nor shall concrete be moved from one part of the work to another by means of the vibrators.
 - (iii) The duration of vibration shall be limited to that required to produce satisfactory compaction without causing segregation. Vibration shall on no account be continued after water or excess grout has appeared on the surface.

- (s) Attendance of Steel Fixer and Carpenter
 - (i) The Contractor shall take adequate steps to ensure that reinforcement, ducts or duct forming devices to house prestressing tendons, formwork and all embedded items are kept in their correct position as concreting proceeds.
 - (ii) The Contractor's arrangements for concreting shall include for a competent steel fixer and a carpenter to be in attendance on the concreting gang as and when required.
- (t) Curing of Concrete
 - (i) Concrete shall be cured by being kept continuously moist throughout the curing period and by protecting the newly cast surface from the effects of sunshine, drying winds, rain, running water or mechanical damage.
 - (ii) The curing shall be maintained for a continuous period of at least:
 - 1) 7 days when the cement used in the concrete is ordinary Portland cement; and
 - 2) 3 days when the cement used in the concrete is rapid hardening cement.
 - (iii) The protection shall be applied as soon as practicable after completion of placing and shall include one or more of the following methods as may best suit the circumstances:
 - 1) By water sprays in continuous operation;
 - 2) Ny covering with hessian or similar absorbent material, or sand, kept constantly wet;
 - 3) After thorough wetting, by covering with a layer of waterproof fabric kept in contact with the concrete surface; and
 - 4) By the application of an approved coloured non-staining liquid curing membrane which is either self-removing or easily removed following the curing period and which has a 75% moisture retention standard. The liquid shall be applied to formed surfaces immediately after stripping the formwork.
 - (iv) Liquid curing membranes shall not be used on Class U 1 surfaces, or where laitance is to be removed and aggregate exposed to provide a satisfactory bond for placing further concrete or mortar screeds, or where in the opinion of the S.O. the use of such a membrane is likely to spoil the finished appearance of an exposed surface.
 - (v) The S.O. may also prohibit the use of liquid curing membranes where their effect on any retained liquid would make them unsuitable as curing agents.
 - (vi) Any concrete which, due to the Contractor's failure to comply with requirements for curing, is in the opinion of the S.O. likely to have been impaired shall forthwith be removed from the Site and replaced.
- (u) Construction Joints
 - (i) A construction joint is defined as a joint in the concrete introduced for convenience in construction at which special measures are taken to achieve subsequent continuity without provision for relative movement.

- (ii) Construction joints shall be located so as not to impair the strength of the concrete. Rebates, keys, notches or waterstops shall not normally be used in construction joints unless otherwise directed by the S.O..
 - (iii) The position of construction joints and the size of the formwork panels shall be so co-ordinated that where possible the line of any construction joint coincides with the line of a formwork joint and that in any case all construction joint lines and formwork joint lines appear as a regular and uniform series.
 - (iv) For all exposed horizontal joints and purposely inclined joints, a uniform joint shall be formed with a batten of approved dimensions to give a straight and neat joint line.
 - (v) Concrete placed to form the face of a construction joint shall have all laitance removed and the large aggregate exposed prior to the placing of fresh concrete. The laitance shall wherever practicable be removed when the concrete has set but not hardened by spraying the concrete surface with water under pressure or brushing with a wire brush sufficient to remove the outer mortar skin and expose the large aggregate without disturbing it. Where the laitance cannot be removed due to hardening of the concrete, the whole of the concrete surface forming the joint shall be treated by high pressure water jet, sand blasting, use of a needle gun or a scaling hammer to remove the surface laitance. Before concreting is resumed, all loose matter on the existing concrete surface shall be removed and the surface slightly wetted.
 - (vi) Concrete shall not be allowed to run to a feather-edge and any vertical joints shall be formed against a stop-end. If a kicker or stub-end is used it shall be at least 70 mm high.
 - (vii) No concreting shall be started until the S.O. has approved the positions and form of the construction joints.
- (v) Movement Joints
- (i) Movement joints are defined as all joints intended to accommodate relative movement between adjoining parts of a structure, special provision being made where necessary for maintaining the watertightness of the joint. The Contractor shall pay due regard to the instructions of manufacturers of proprietary jointing materials and shall, if required by the S.O., demonstrate that the jointing materials can be applied satisfactorily.
 - (ii) As far as possible jointing of waterstops on Site shall be confined to the making of butt joints in straight runs. Where it is necessary for an intersection, or change of direction, or any joint other than a butt joint in a straight run, to be made on Site, a preliminary joint, intersection or change of direction piece shall be made and submitted to such tests as the S.O. may require.
 - (iii) Flexible waterstops shall be fully supported in the formwork, free of nails and clear of reinforcement and other fixtures. Any waterstop which after installation is found to be damaged shall be removed and replaced. Concreting in the vicinity of waterstops shall be carried out with care to ensure that the water stops do not bend or distort while the concrete is being placed and compacted in position.

- (iv) The surface of set concrete in a movement joint shall, where specified on the Drawings, be painted with two coats of bituminous paint and new concrete shall be placed against it only when the paint is dry. Expansion joints shall be formed by a separating strip of approved preformed joint filler.
- (v) Caulking grooves shall be provided as shown on the Drawings. At all joints where a caulking groove is formed, the groove shall be wire brushed and loose material removed and blown out by compressed air immediately prior to caulking. After the groove has dried it shall be primed and caulked with approved jointing compound. At all caulked joints, the face of the caulking strip and 50 mm width of concrete on either side shall be painted with two coats of paint having the same base as the caulking compound.
- (w) Dimensions and Surfaces of Finished Concrete
 - (i) Workmanship in formwork and concreting shall be such that finished concrete surfaces be seen as smooth and tidy which should not require any further treatment to rectify the undulations. Concrete surfaces for the various classes of unformed and formed finishes shall in any event never exceed the maximum permitted tolerances stated in the Specification or Drawings.
 - (ii) In the table 'line and level' and 'dimension' shall mean the lines, levels and cross-sectional dimensions shown on the Drawings.
 - (iii) Surface irregularities shall be classified as 'abrupt' or 'gradual'. Abrupt irregularities include, but shall not be limited to, offsets and fins caused by displaced or misplaced formwork, loose knots and other defects in formwork materials, and shall be tested by direct measurement. Gradual irregularities shall be tested by means of a straight template for plane surfaces or its suitable equivalent for curved surfaces, the template being 3.0 m long for unformed surfaces and 1.5 m long for formed surfaces.
 - (iv) The class of finish for unformed surfaces is described below. The class of finish for formed surfaces is described in the Formwork Specification.

Maximum tolerance (mm) in:				
Class of finish	Line and level	Abrupt irregularity	Gradual irregularity	Dimension
U1	± 12	6	± 6	-
U2	± 6	3	± 3	-
U3	± 6	3	± 3	-
F1	± 12	6	± 6	+ 12, - 6
F2	± 6	6	± 6	+ 12, - 6
F3	± 3	3	± 3	± 6

(x) Unformed Surfaces - Class of Finish

- (i) Finishes to unformed surfaces of concrete shall be classified as U1, U2, U3, 'spaded' or 'bonded concrete' or such other special finish as may be particularly specified. Where the class of finish is not specified the concrete shall be finished to Class U1 if the concrete is to be buried on completion of the Permanent Works and shall be finished to Class U3 if the concrete is to be exposed on completion of the Permanent Works.
- (ii) Class U1 finish is the first stage for Class U2 and U3 finishes and for a bonded concrete surface. Class U1 finish shall be a levelled and screeded, uniform plain or ridged finish, which (unless it is being converted to Class U2, U3 or bonded concrete) shall not be disturbed in any way after the initial set and during the period of curing, surplus concrete being struck off immediately after compaction.
- (iii) Where a bonded concrete surface is specified, the laitance shall be removed from the Class U1 finish surface and the aggregate exposed while the concrete is still green.
- (iv) A spaded finish shall be a surface free from voids and brought to a reasonably uniform appearance by the use of shovels as it is placed in the Works.
- (v) Class U2 finish shall be produced by manual or mechanical floating of the concrete surface after the initial set has taken place and the surface has hardened sufficiently. The concrete shall be worked no more than is necessary to produce a uniform 'sandpaper' finish free from screed marks.
- (vi) Class U3 finish shall be a hard smooth steel-trowelled finish. Trowelling shall not commence until the moisture film has disappeared and the concrete has hardened sufficiently to prevent excess laitance from being worked into the surface. The surface shall be trowelled under firm pressure and left free from trowel marks.
- (vii) Cement, mortar or water shall not be added during any of the above operations.
- (viii) Unless otherwise specified or shown on the Drawings, unformed finishes U1, U2 or U3 shall be used at the following locations:
 - 1) Concrete surfaces upon which an applied finish (excluding concrete surface, protective membrane hardener) is subsequently to be placed such as screed, tiling, granolithic topping etc;
 - 2) Exposed surfaces external to buildings and within tanks; and
 - 3) Concrete surfaces within buildings and other structures upon which an applied finish other than concrete surface hardener is not subsequently to be applied

(y) Building in Pipes and Other Items

- (i) All pipes and other items passing through concrete structures shall wherever practicable be built into the structure as work proceeds, having been installed and connected to the remainder of the system to ensure proper fit before the start of any concreting.

- (ii) All items to be built in shall be securely supported in their correct position so as to prevent movement or damage during building-in. No pipe with flanged joints shall be concreted in until its accurate fit with other pipework has been checked and it has been secured in position.
- (iii) Complete sections of pipework, including the items to be built into structures, shall be erected and checked by the Contractor for line and level before any items are built in. Casting in of individual pipes in isolation of the rest of the system will not normally be permitted, and any proposals in this respect shall be subject to the approval of the S.O..
- (iv) The Contractor shall erect and fix securely in position all pipework, which is to be built in. Any supports shall be clear of the openings into which pipes are to be built, to allow access for installation of formwork and placing concrete.
- (v) The Contractor shall recheck the pipework alignment after erection of formwork and before placing concrete starts, to ensure that it has not moved.
- (vi) The Contractor shall be responsible for ensuring that the pipes are built in the correct position and true to line and level and special care shall be taken to ensure that concrete is adequately worked and placed around the pipe to secure a completely watertight joint.
- (vii) Particular care shall be taken to ensure that no pipe is displaced or disturbed in alignment whilst this operation is carried out. If pipes are found to have moved, the Contractor shall be required to cut out such pipes and re-fix them true to line and level at his own expense.
- (viii) The Contractor may apply to the S.O. to install pipework or other cast in features using boxouts. The boxout edges shall be well scabbled and wedge shaped with the narrowest opening located on the water retaining side. The edges of the boxout shall be square to the face of the wall or slab for a minimum of 25 mm to avoid feather edges. The boxout shall have a 100 mm by 100 mm key groove to all sides at the centre of the wall or slab. Anti crack steel shall be exposed at the corners of the boxout to improve the box out bond. The Contractor shall use a low water cement ratio ultra high slump non-shrink concrete mix to cast in the boxout. The Contractor shall consider the use of 'No Compaction Concrete' to achieve the high level of compaction required. The Contractor shall submit for the approval of the S.O. the following details:
 - 1) Any proposed modifications to the steel reinforcement design including structural calculations, if required;
 - 2) The size and shape of the proposed box out; and
 - 3) Methodology of ensuring good bond and key between new concrete used to fill boxout.
- (ix) Methodology for Supporting Pipe or Fitting during Casting In
 - 1) Mix design for concrete used to cast fitting in place. Mix shall ensure minimum shrinkage; and
 - 2) Methodology proposed to pour and compact concrete for casting fitting in place.

(z) Structural Precast Concrete

- (i) Structural units of concrete shall be manufactured in the grade of concrete and to the sizes and details shown on the Drawings. The concrete shall comply with the provisions of the Specification whether such units are manufactured on the Site or obtained from manufacturers approved by the S.O..
- (ii) Where the installation of precast concrete units in any structure is such that the faces of the units are to be left exposed either internally or externally, the exposed surfaces of the units as finished shall be uniform in colour and in texture. All cement, aggregates and other materials used in the manufacture of the units shall be obtained from the same approved sources throughout the period of manufacture.
- (iii) Formwork and unformed surfaces for precast concrete units shall comply with the requirements for F3 formed surfaces and U3 unformed surfaces respectively.
- (iv) The Contractor shall submit full details of his proposed method of carrying out all operations connected with the manufacture and assembly of precast concrete structural members, including:
 - 1) A description of the types of casting bed, mould and formwork for the various types of members;
 - 2) The procedure for concrete casting and the method of curing the concrete;
 - 3) The procedure for transporting, handling, hoisting and placing of each type of precast structural member; and
 - 4) Particulars of the temporary supports deemed necessary to ensure adequate stability during erection and fully proof against the effects of construction loads, wind load and other transient loads.

(aa) Dimensional Tolerances

- (i) Dimensions and shapes of precast concrete structural members and for in site concrete shall comply with tolerances given on the Drawings or, when not so given, with those stated above.

(bb) Installation of Precast Concrete Structural Members

- (i) At all stages and until completion of the Works, precast members shall be adequately protected to preserve all permanently exposed surfaces, arises and architectural features. The protection shall not mark or otherwise disfigure the concrete.
- (ii) All units shall be laid, bedded, jointed and fixed in accordance with the lines, levels and other details shown on the Drawings.
- (iii) Dry-pack mortar jointing for packing shall consist of one part by volume of ordinary Portland cement and two parts by volume of fine aggregate passing a 1 mm sieve. The mortar shall be mixed with only sufficient water to make the materials stick together when being moulded in the hands. The mortar shall be placed and packed in stages, where possible from both sides of the space being filled, using a hardwood stick hammered until the mortar is thoroughly compacted.

- (iv) Any precast concrete structural member which is found to be cracked, damaged or otherwise inferior in quality, either before or after erection, shall be rejected or replaced by the Contractor.
- (v) The Contractor shall submit for approval his proposals for the following:
 - 1) Minimum ages for handling and transportation;
 - 2) Position and design of lifting points;
 - 3) Method of lifting;
 - 4) Lifting equipment;
 - 5) Methods of supporting and stacking;
 - 6) Methods of assembly and erection;
 - 7) Accuracy of assembly and erection;
 - 8) Temporary support;
 - 9) Final structural connections;
 - 10) Method of protecting units from damage;
 - 11) Marking for identification;
 - 12) Preparation of contact surfaces; and
 - 13) Removal of locating devices after erection.
- (cc) Installation of Precast Concrete Units
 - (i) At all stages and until completion of the work, precast members shall be adequately protected to preserve all permanently exposed surfaces, arises and architectural features. The protection shall not mark or otherwise disfigure the concrete.
 - (ii) All units shall be laid, bedded, jointed and fixed in accordance with the lines, levels and other details shown on the Drawings.
 - (iii) Dry pack mortar jointing for packing shall consist of one part by volume ordinary Portland cement and two and half parts by volume of fine aggregate passing a 1 mm sieve. The mortar shall be mixed with only sufficient water to make the materials stick together when being moulded in the hands. The mortar shall be placed and packed in stages, wherever possible from both sides of the space being filled, using a hardwood stick hammered until the mortar is thoroughly compacted.
 - (iv) Any precast concrete structural member which is found to be cracked, damaged or otherwise inferior in quality either before or after erection, shall be rejected and replaced by the Contractor.
- (dd) Precast Concrete Products
 - (i) Precast concrete products shall where appropriate be constructed in accordance with the relevant standards.
 - (ii) The concrete shall comply in every aspect with the provisions of the Specification whether such products are manufactured on the Site or obtained from other manufacturers.

(ee) Prestressed Concrete

- (i) Prestressed concrete work whether precast or cast in situ shall be as detailed on the Drawings and the Contractor shall provide full details of his proposed method of carrying out all operations connected with prestressing including:
 - 1) Design of concrete mix;
 - 2) Description of concrete casting and curing operations;
 - 3) Procedure for prestressing and method of checking and recording the prestress, wire extension and anchorage slip;
 - 4) Procedure for handling, hoisting and placing of any precast units;
 - 5) Grouting of ducts;
 - 6) All tendons, duct forming devices, anchorages and other components shall be kept free of mud, oil (except water soluble oil applied for protection), paint, retarders, loose rust, or other foreign matter. They shall be placed with a tolerance of ± 3 mm in concrete dimensions of 300 mm or less or ± 6 mm in concrete dimensions over 300 mm; and
 - 7) The bearing surfaces between anchorages and concrete shall be normal to and concentric with the tendons and the line of action of the prestressing force.

(ff) Machinery Bases and Grouting in

- (i) Bases to take machinery and associated pipework shall be constructed in fair faced concrete to the dimensions shown on the approved machinery drawings or as ordered by the S.O.. The mounting surface of the base shall be steel floated to true and level planes.
- (ii) The structural concrete on which the bases are to be erected shall be prepared by hacking and cleaning off.
- (iii) Bases shall be tied to the structural concrete with vertical reinforcement unless otherwise detailed. Horizontal reinforcement shall also normally be provided at the level of the pockets for the holding down bolts.
- (iv) Bolt pockets and lead ins for grout shall be formed by a means which shall have the approval of the S.O.. Polystyrene formers shall not be used. Where expanded metal is used as a former it shall be left in.
- (v) When the machinery has been erected the bolt pockets shall on approval of the S.O. be completely filled using a mixture of 2 parts of cement to 3 parts sharp sand by volume with sufficient water to form a thick creamy consistency.
- (vi) The machinery shall be run under the supervision of the Contractor and witnessed by the S.O. after the grout has hardened. When so directed the Contractor shall complete the grouting operation by filling the space between the top of the concrete and the underside of the machinery base plate.

G3.20.6 Testing

(a) Sampling and Testing of Aggregates

- (i) The Contractor shall take samples of all aggregates and test them for grading, by the methods described in SS EN 12620, at least once in each week when concreting is in progress and additionally as the S.O. may require.
- (ii) Whenever the source of aggregate is changed and in any case at least once per month the Contractor shall carry out tests on aggregates relating to water absorption, potential alkali reaction, chloride content, sulphate content and shrinkage characteristics, all as specified.

(b) Sampling and Testing of Concrete

- (i) The Contractor shall provide the necessary equipment and shall determine the compacting factor of the freshly mixed concrete by the method described in BS EN 12350 on each occasion that a set of test cubes is made and at such other times as the S.O. may direct.
- (ii) For each grade of concrete (except grade 10) works test cubes shall be made whenever required by the S.O. but not less frequently than as follows unless otherwise particularly specified and/or agreed with the S.O..
 - 1) For prestressed concrete members or for structural reinforced concrete members; and

Vol. of concrete (m ³)	Minimum numbers of samples
< 10	1
11 – 40	2
41 – 100	3
101 – 400	3 + 1 sample per 100m ³ or less
401 – 2500	6 + 1 sample per 200m ³ or less
> 2500	17 + 1 sample per 250m ³ or less

- 2) For mass concrete of any grade required for each range of the concrete volume the number of samples in item (1) shall be reduced proportionally by half and rounded up to the next highest integer.
- (iii) The actual rate of sampling at any time may be expected to fluctuate according to the random selection of batches to be sampled and may be increased in appropriate circumstances.
- (iv) A batch shall be defined as the quantity of concrete mixed in one cycle of operations of a batch mixer or the quantity of concrete delivered to site ready mixed in a vehicle.
- (v) Where there are more than one plant supply, the sampling rate shall be proportionally distributed among the suppliers based on the volume of concrete generated from the supplier's plant.
- (vi) An early assessment shall be made of the 28-day strength using a method acceptable to the Engineer. The mix shall be adjusted as necessary to achieve the specific characteristic strength.

- (vii) For large pours the sampling rate may be reduced with the approval of the S.O..
 - (viii) Each set of cubes (Four cubes per set) shall be made from a single sample taken from a randomly selected batch of concrete. Two cubes shall be tested 7 days after manufacture and two cubes 28 days after manufacture. The average of the two 28 day results shall be taken as the test result.
 - (ix) Where concrete of a particular grade is likely to be placed infrequently, and the above rates of sampling might not produce sufficient representative test cubes to enable the concrete quality to be monitored properly, the Contractor shall increase the number of standard samples taken during each day when concrete of that grade is being placed so as to ensure that enough representative test cubes are obtained.
 - (x) For large, continuous concrete pours such as for slabs, decks etc., the number of test cubes may be reduced subject to the prior, approval of the S.O., in accordance with the recommendations of BS 5328 and BS EN 12350.
- (c) Compliance with Specified Requirements
- (i) The concrete shall be deemed satisfactory only if the conditions given in both (1) and (2) are strictly met:
 - 1) The average 28 day strength determined from 4 consecutive test results exceeds the specified characteristic strength by not less than 2 N/mm² for grade 15 concrete and 3 N/mm² for grades 20 and above; and
 - 2) Each individual test result is greater than 90% of the specified characteristic strength.
 - (ii) If only one cube result fails to meet the requirements of (2) above, that result may be considered to represent only the particular batch of concrete from which the cube was taken. That particular group will then be deemed to have failed.
 - (iii) If more than one cube in a group fails to meet the requirement of (2) above, or if the average strength of any group of four consecutive test cubes fails to meet the requirement of (1) above, then all the concrete in all the batches represented by such cubes shall be deemed not to meet the strength requirements.
- (d) Action in the Event of Failure to Meet Requirements
- (i) When concrete of a particular design of mix does not meet the strength requirements set out in this specification, no further concrete from that mix shall be placed in the Works and the Contractor shall establish the cause of the failure and apply such remedies as are necessary. The Contractor shall revise the mix accordingly and demonstrate by trial mixes and test cube results that the revised mix meets the specified requirements.
 - (ii) The Contractor shall within 24 hours of the date of test make proposals for approval by the S.O. for action to be taken in respect of any concrete represented by test cubes which fail to meet the foregoing requirements. These proposals may include, but shall not be limited to, cutting and testing cores. In the absence of such approved proposals the S.O. will instruct the Contractor regarding the action to be taken.

- (e) Cutting and Testing of Core Samples
 - (i) For the examination and testing of hardened concrete, cylindrical core specimens of 150 mm nominal diameter shall be cut from it as and where directed by the S.O.. They shall be cut perpendicular to the face. The preferred ratio of length to diameter shall be 2. The procedure for drilling, examination, measurement and testing for compressive strength shall be in conformity with good practice. Before preparation for testing, the specimen shall be made available for examination by the S.O.. If the estimated in-situ cube strength of the specimen determined in accordance with BS EN 12390 is less than the specified characteristic strength at 28 days, or if in the opinion of the S.O. the concrete fails to meet the Specification in other respects, the concrete in that part of the Works of which it is a sample shall be deemed not to comply with the Specification.
- (f) Action in the Event of Concrete Not Complying
 - (i) Concrete which is found not to comply with the Specification shall be rejected and the Contractor shall submit to the S.O. for his approval a method statement detailing how the concrete shall be broken out and replaced.
- (g) Contractor Responsible for Testing
 - (i) In addition to any specific obligations for sampling and testing the Contractor shall be responsible for routine inspection sampling and testing of concrete, materials, measuring devices and plant, in order to control the quality of work and to ensure compliance with the Specification including conformity with approved samples.
 - (ii) The Contractor will be allowed reasonable opportunity for the purposes of quality control to use the laboratory and testing equipment provided under the Contract for the use of the S.O. and his staff.

G3.20.7 Concrete Jointing and Protection

- (a) Definitions
 - (i) The following terms shall have the meaning hereby assigned to them:

Construction Joint	A joint in the concrete introduced for convenience in construction at which special measures are taken to achieve subsequent continuity without provision for further relative movement
Movement Joint	A joint in the concrete intended to accommodate relative movement between adjoining parts of a structure
Expansion Joint	A movement joint intended to accommodate expansion and contraction of the concrete without restraint
Contraction Joint	A movement joint intended to accommodate contraction of the concrete without constraint, in the case of a complete contraction joint, or with some constraint, in the case of a partial contraction joint

Induced Contraction Joint	A contraction joint formed not by using a temporary stop end but by reducing the depth of the concrete section by at least 25%
Movement Accommodation Factor	The maximum movement that a sealant is capable of tolerating throughout its working life, expressed as a percentage of the joint width

(b) Construction Joints

- (i) Construction joints shall be located so as not to impair the strength of the concrete. Rebates, keys, notches or waterstops shall not normally be used in construction joints unless otherwise directed by the S.O..
- (ii) The position of construction joints and the size of the formwork panels shall be so co-ordinated that where possible the line of any construction joint coincides with the line of a formwork joint and that in any case all construction joint lines and formwork joint lines appear as a regular and uniform series.
- (iii) For all exposed horizontal joints and purposely inclined joints, a uniform joint shall be formed with a batten of approved dimensions to give a straight and neat joint line.
- (iv) Concrete placed to form the face of a construction joint shall have all laitance removed and the large aggregate exposed prior to the placing of fresh concrete. The laitance shall wherever practicable be removed when the concrete has set but not hardened by spraying the concrete surface with water under pressure or brushing with a wire brush sufficient to remove the outer mortar skin and expose the large aggregate without disturbing it. Where the laitance cannot be removed due to hardening of the concrete, the whole of the concrete surface forming the joint shall be treated by high pressure water jet, sand blasting, use of a needle gun or a scaling hammer to remove the surface laitance. Before concreting is resumed, all loose matter on the existing concrete surface shall be removed and the surface slightly wetted.
- (v) Concrete shall not be allowed to run to a feather-edge and any vertical joints shall be formed against a stop-end. If a kicker or stub-end is used it shall be at least 70 mm high.
- (vi) No concreting shall be started until the S.O. has approved the positions and form of the construction joints.

(c) Movement Joints

- (i) Movement joints are defined as all joints intended to accommodate relative movement between adjoining parts of a structure, special provision being made where necessary for maintaining the water tightness of the joint. The Contractor shall pay due regard to the instructions of manufacturers of proprietary jointing materials and shall, if required by the S.O., demonstrate that the jointing materials can be applied satisfactorily.

- (ii) As far as possible jointing of waterstops on Site shall be confined to the making of butt joints in straight runs. Where it is necessary for an intersection, or change of direction, or any joint other than a butt joint in a straight run, to be made on Site, a preliminary joint, intersection or change of direction piece shall be made and submitted to such tests as the S.O. may require.
- (iii) Flexible waterstops shall be fully supported in the formwork, free of nails and clear of reinforcement and other fixtures. Any waterstops which after installation is found to be damaged shall be removed and replaced. Concreting in the vicinity of waterstops shall be carried out with care to ensure that the water stops do not bend or distort while the concrete is being placed and compacted in position.
- (iv) The surface of set concrete in a movement joint shall, be painted with two coats of bituminous paint and new concrete shall be placed against it only when the paint is dry. Expansion joints shall be formed by a separating strip of approved preformed joint filler.
- (v) Caulking grooves shall be provided as shown on the Drawings. At all joints where a caulking groove is formed, the groove shall be wire brushed and loose material removed and blown out by compressed air immediately prior to caulking. After the groove has dried it shall be primed and caulked with bituminous joint sealant. At all caulked joints, the face of the caulking strip and 50 mm width of concrete on either side shall be painted with two coats of paint having the same base as the caulking compound.
- (d) Reference Standards for Concrete Jointing and Protection
 - (i) Unless otherwise specified, concrete jointing and protection shall comply with the relevant Singapore Standards and the following Reference Standards, where applicable:

Standard	Subject
BS 2499	Hot applied joint sealant systems for concrete pavements
BS 3416	Bitumen-based coatings for cold application, suitable for use in contact with potable water
BS 5215	One-part gun grade polysulphide-based sealants
BS 6213	Guide to selection of construction sealants
BS 6949	Bitumen-based coatings for cold application excluding use in contact with potable water
BS 8102	Protection of structures against water from the ground
ASTM D 1751	Preformed expansion joint filler for concrete paving and structural construction (non-extruding and resilient bituminous types)
ASTM D 1752	Preformed sponge rubber and cork expansion joint fillers for concrete paving and structural construction
ASTM D 545	Testing preformed expansion joint fillers for concrete construction (non-extruding and resilient types)

- (e) Submissions by the Contractor for Concrete Jointing and Protection.
 - (i) Submissions which the Contractor is required to make in relation to concrete jointing and protection include, where relevant, the following:
 - 1) Data
 - Types of material and their properties;
 - Manufacturers' catalogues;
 - Manufacturers' proposals for choice and application of products with details of the joint geometry where applicable and the allowable movement accommodation factor; and
 - Manufacturers' recommendations for storage handling, and installation of products, including where relevant:
 - Extent of allowable movements of joints;
 - Design life; and
 - Method of fixing during installation.
 - 2) Drawings
 - Full-scale dimensioned drawings of waterstop cross sections.
 - 3) Samples
 - Where required by the S.O., samples of materials for inspection or independent testing or both.
 - 4) Certificates
 - Manufacturers' and suppliers' certificates of compliance with relevant standards. Such certificates shall state the results of tests carried out.
- (f) Materials
 - (i) Materials Generally
 - 1) Materials for concrete jointing and protection shall be obtained from manufacturers approved by the S.O.. Unless otherwise approved, all materials for concrete jointing and protection which are in contact with each other or jointed together to form a jointing and/or protective system shall be obtained from a single manufacturer.
 - 2) All materials for concrete jointing and protection which may come into contact with one another shall be compatible with each other. They shall not cause any change in properties of the materials with which they are in contact which may make them unsuitable for use in concrete jointing or protection, or which may reduce the working life of joints or protection.
 - 3) The Contractor shall supply manufacturers' test sheets to certify compliance of all jointing and protection materials with relevant quality standards. Where there is no relevant quality standard, the Contractor shall prove by demonstration, tests or otherwise the suitability, adequacy and performance of the material under Site conditions. Only such materials as have been approved by the S.O. shall be used in the Works.

(ii) Materials in Contact with Potable Water

- 1) No substance or product which could impart toxicity, taste, odour or colour to water or be otherwise objectionable on health ground shall be used in the Works where it may be applied to or introduced into water which is to be supplied for drinking, washing or cooking.
- 2) Pourable joint fillers for potable water structures shall be NSF approved or by the governing health agency.

(iii) Materials in Contact with Sewage

- 1) Materials which may come into contact with sewage shall not be susceptible to chemical or biological attack, and shall retain their specified properties.

(iv) Waterstop

- 1) Unless specified otherwise on the Drawings or elsewhere in the specification waterstop shall be of the following types:
 - Rubber or PVC waterstop for use within the thickness of the concrete member; and
 - PVC waterstop for use on the outer face of the concrete member.
- 2) Waterstop shall be dense, Homogeneous and free of holes or other imperfections. The cross section shall be uniform along the length and symmetrical transversely so that the thickness at any given distance from either edge will be uniform. Where not otherwise shown on the Drawings:
 - The width of the waterstop shall be not less than 80% of the thickness of the concrete section and not less than 250 mm;
 - The web thickness shall be not less than 9 mm in the case of waterstop for use within the concrete thickness or 5 mm in the case of waterstop for use on the outer face;
 - All internal waterstop shall incorporate a solid bulb or major rib at or near each edge;
 - Waterstop wider than 180 mm which is for use on the outer face shall incorporate a total of at least two major ribs each side of the centreline;
 - Waterstop at movement joints shall incorporate a central hollow bulb;
 - Waterstop for use at induced contraction joints shall incorporate a crack inducer; and
 - Waterstop for use within the concrete thickness shall be of rubber.
- 3) Rubber waterstop shall be of natural rubber, suitable synthetic rubber, or a blend of the two. PVC waterstop shall be of an elastomeric plastic compound; no reground material shall be used.

- 4) Waterstop shall have the following properties when tested at 25°C.

Property	Rubber	PVC
Minimum tensile strength (N/mm ²)	18	13.75
Minimum elongation at break (%)	450	280
Hardness (IRHD/Shore 'A')	60 - 70	70 - 90
Softness (BS 2571)	-	42° - 52°
Specific gravity	1.05 - 1.15	-

- 5) All intersections and junction in waterstop shall be factory-produced by the manufacturer of the waterstop, normally as moulded special pieces, so that the only joints remaining to be made at Site shall be straight butt joints between straight lengths of waterstop of the same section and material.

(v) Joint Filler and Sealant

- 1) Joint fillers shall be preformed, compressible and resilient and shall comply with all relevant requirements of this clause, and should be non-rotting and non-absorbent material except where alternative requirements are specified, shown on the Drawings or specifically approved by the S.O..
- 2) Bitumen impregnated joint fillers of fibre or cork shall comply with ASTM D 1751.
- 3) Resin bonded cork or self-expanding cork joint filler shall comply with ASTM D 1752. Sponge rubber joint fillers shall also comply with ASTM D 1752 except that the load required to compress the test specimen to 50% of its initial thickness shall not exceed 0.5 N/mm².
- 4) Joint fillers of closed-cell polyethylene or polyvinyl foam shall, when tested in accordance with ASTM D 545, have the following properties:
- 5) The load required to compress the test specimen to 50% of its thickness before test shall be no less than 0.13 N/mm² nor greater than 0.3 N/mm²
- 6) 10 minutes after release of the compressive load (or 1 hour if the retest provision applies) the test specimen shall have recovered not less than 80% of its initial thickness.
- 7) Joint filler for use in joints which may be submerged shall be of closed-cell polyethylene or self-expanding cork.

(vi) Joint Sealant

- 1) Joint sealant shall be one of the following types as shown on the Drawings or specified elsewhere:
 - Polysulphide-based sealant, having a movement accommodation factor of at least 25% and a minimum service life expectancy of 15 years;

- Polysulphide-based sealant, having a movement accommodation factor of at least 10% and a minimum service life expectancy of 15 years;
 - Approved proprietary product formulated from chemically resistant elastomeric polymers, having a movement accommodation factor of at least 20% and a minimum life expectancy of 15 years;
 - Rubber-bitumen compound, type A2, having a movement accommodation factor of at least 10% and a minimum service life expectancy of 10 years; and
 - Approved bituminous putty proprietary product, having a movement accommodation factor of at least 10% and a minimum life expectancy of 10 years.
- 2) Sealant shall be chemically resistant non-degradable under the prevailing conditions to which the sealant may be exposed. Where sealant will be in contact with sewage, the Contractor shall submit for the S.O.'s approval evidence that the proposed product will resist aerobic and anaerobic attack.
- 3) Where specified, fire resistant sealant shall resist the effects of heat and fire without significant reduction in integrity or insulation for the required duration. Fire resistant sealants shall be certified by the Singapore's Fire Safety Bureau or the relevant local authority.

(vii) Back-up Material, Bond Breaker and Primer

- 1) Back-up material inserted in a joint to limit the depth of sealant (where joint filler does not serve this purpose) shall comprise closed-cell low-density compressible joint filler preformed for the purpose or such other material as is recommended for this use by the sealant manufacturer and approved by the S.O..
- 2) Bond breaker installed behind joint sealant shall be thin self-adhesive polyethylene tape. Where the sealant manufacturer recommends the use of primer before the application of sealant, it shall be provided by the sealant manufacturer for the purpose.

(viii) Slip Membrane

- Slip membrane for use in a two-layer separating membrane at sliding joints shall consist of plastic sheet not less than 1.5 mm thick which shall have a coefficient of friction not exceeding 0.2 between rubbing surfaces when used in accordance with the manufacturer's instructions and subjected to a load of 0.27 N/m².

(ix) Bituminous Paint

- 1) Bond breaking paint for use on concrete or steel reinforcement shall be bituminous paint.

(x) Protective Membrane Systems

- 1) Membranes for the exclusion of groundwater or the retaining of an aqueous liquid shall comply with the relevant standards and shall be flexible self-adhesive impervious composite sheeting of total thickness not less than 1.5 mm consisting of either:
 - A sheet not less than 0.30 mm thick of cross-laminated polyethylene or PVC, and a rubber-bitumen compound; or
 - A polyester-reinforced rubber-bitumen sheet with polyester carrier film.
- 2) Primer for membrane and other sundry materials for forming fillets, upstands and chamfers shall be fully compatible with the membrane material.

(g) Workmanship for Concrete Jointing and Protection

(i) Storage of Materials

- 1) All materials for concrete jointing and protection shall be stored under weatherproof cover. Any storage conditions recommended by the manufacturer of any materials for concrete jointing and protection shall be observed.
- 2) Materials shall be used in order of delivery from the manufacturer and shall not be used after the expiry of any storage life recommended by the manufacturer.

(ii) Joint Filler

- 1) Joint fillers employed to form and fill expansion joints shall have the same nominal thickness as the joint width specified or shown on the Drawings and shall extend through the full thickness of the joint not occupied by other jointing materials.
- 2) The joint filler shall be fixed by dabs of adhesive to the surface of the first placed concrete forming one side of the joint. Alternatively, the joint filler shall be placed against the formwork on the concrete side of the first pour. The latter method shall be used when the joint filler edge is to be finished with capping strip. In that case, joints between lengths of capping strip shall be made by heat fusion and the capping strip pushed firmly over the edge of the joint filler which shall be accurately pre-cut.
- 3) Where sealant is required, the recess or caulking groove to receive it may be formed by the use of removable forms or where appropriate by cutting back the filler to the required depth, the filler being incised upon installation to facilitate this.

(iii) Joint Sealing

- 1) The caulking groove to accept sealant shall be formed to the dimensions shown on the Drawings or recommended by the sealant manufacturer and approved by the S.O.. After the concrete has cured and dried out, the bonding faces of the caulking groove shall be first grit blasted or wire brushed and then vacuum cleaned to remove all dirt, surface laitance and dust. The adjacent areas shall also be thoroughly vacuum cleaned to reduce the risk of cleaned caulking grooves becoming contaminated.
- 2) Any wet or damp areas shall be allowed to dry out, or where necessary shall be dried out, before joint sealing materials are applied.
- 3) Where back-up material is required it shall be at least 20% wider than the joint and shall be placed accurately in position to define the required depth of sealant.
- 4) The back of the caulking groove, whether of concrete or joint filler shall be covered by bond breaker tape except where the joint filler is cellular polyethylene or where the caulking groove forms part of a structural hinge joint.
- 5) The sides of the caulking groove shall be primed where appropriate in accordance with the joint sealant manufacturer's instructions.
- 6) On permanently exposed areas of structures, joint sealing is to be carried out with the aid of masking tape to form neatly defined surface limits to the seal.
- 7) Should sealant suffer any of the types of failure, it shall be replaced after the possible causes of failure have been investigated and the appropriate remedial action has been taken.

(iv) Slip Membrane

- 1) Surfaces to receive slip membrane shall be smooth, plane and clean. Concrete surfaces shall be Class U3 finish.
- 2) Two layers of slip membrane shall be cut to size and fastened together along all edges using masking tape. The complete assembly shall then be placed and fixed in position in accordance with the manufacturer's instructions.

(v) Bituminous Painting

- 1) Where specified or shown on the Drawings, concrete surfaces or reinforcement shall be painted with two coats of bituminous paint. Concrete surfaces to be painted shall be of Class F2 finish, smooth, plane, clean and dry. New concrete shall not be placed against the painted surface until the paint is dry.

(vi) Protective Membrane

- 1) All concrete to which protective membrane is to be applied shall be of Class U2 or F2 finish, free from sharp protrusions and hollows. External angles shall be chamfered. All surfaces shall be clean and dry. Except for horizontal blinding concrete, all concrete to receive membrane shall be primed in accordance with the membrane manufacturer's instructions.

- 2) One complete layer of protective membrane shall be applied in accordance with the manufacturer's instructions with laps of not less than 150 mm at ends and 75 mm at sides. All corners shall receive two layers of membrane either by use of an additional reinforcing strip of membrane not less than 300 mm wide or by carrying each overlapping membrane at least 150 mm beyond the corner to provide an overlap not less than 300 mm wide. Internal angles shall have approved fillets installed to facilitate proper dressing of the protective membrane with the corner.
- 3) Chamfers and fillets shall be not smaller than 25 mm x 25 mm.

TABLE G3.20-1 PERMISSIBLE DEVIATIONS FOR IN-SITU AND PRECAST CONCRETE CONSTRUCTION			
Item of Construction	Type of Measurements	Elements	Allowable Tolerance (mm)
1 Size and shape of elements and components	a) Cross section dimensions	Beams, columns and slabs	2 per 100 mm or 5 whichever is greater subject to a maximum of +15 or - 5
	b) Length	Precast members	1 per metre or 5 whichever is greater
	c) *Straightness measured in any direction in length of: i) 0.5 m ii) up to 5 m	Columns, walls and beams	10 20
2 Variation in position of elements	a) Departure of any point from its position in plan as shown on the drawings	Columns, beams and walls	±15
		Foundations	±35
	b) Deviation in the mean level with reference to the nearest TBM	Corbels	±5
		Staircase tread	±5 subject to a cumulative total of ± 10 between landings

G3.20.8 Grout and Miscellaneous Concrete Work

- (a) This section Includes:
 - (i) Grouting under base plates for equipment and fittings; and
 - (ii) Miscellaneous concrete work.
- (b) References
 - (i) ACI 351.1R Grouting for Support of Equipment and Machinery; and
 - (ii) CSA Standard A23.1, Grouting for base plates for structural steel beams and column.
- (c) Product Data Sheets
 - (i) The Contractor shall submit three copies of manufacturer's product data sheets including installation, application, and maintenance instructions for:
 - 1) Non shrink non ferrous grout with intended use; and
 - 2) Concrete repair products.
- (d) The Contractor shall engage manufacturer's representative for on site supervision during and after installation where required. The Contractor shall verify that specified products are correctly installed and comply with manufacturer's printed instructions.
- (e) The Contractor shall comply with the following installation requirements as follows:
 - (i) Grouting Under Base Plates for Equipment and Fittings
 - 1) Install grout in accordance with Chapter 6 - Preparation for Grouting, Chapter 7 - Grouting Procedures, and Chapter 8 - Curing and Protection, of ACI 351.1R except as modified or supplemented by the manufacturer;
 - 2) Roughen and clean concrete surface prior to installation of base plates;
 - 3) Use preblended non shrink non ferrous hydraulic cement grout as specified for the application except where equipment supplier specifically recommends non shrink epoxy grout; and
 - 4) After the trades responsible have properly placed, levelled, and aligned base plates, mix and place a minimum of 25mm grout under base plates completely filling the voids.
 - (ii) Grouting Under Structural Steel Base Plates
 - 1) Install grout under all base plates supporting structural steel, beams, columns, etc. after steel erection is completed; and
 - 2) Use preblended non shrink non ferrous hydraulic cement grout as specified.

- (f) Repair of Existing Deteriorated Concrete
 - (i) During shut-downs and tie-ins between new and existing structures, conditions of deteriorated existing concrete, which in the Engineer's opinion will require repairs, may be discovered. In such cases the S.O. may direct the Contractor to carry out repair work in conformance with the requirements. It is expected that such concrete repair work will fall into one of the following categories:
 - 1) Non-structural repairs – surface preparation and patching; and
 - 2) Structural repairs – cutting out unsound concrete repairing corroded reinforcing steel and repair of concrete openings.
 - (ii) Non-Structural Repairs: Work would include concrete surface repairs where depth of concrete deterioration would be less than the concrete cover over the reinforcement:
 - 1) Do not repair defective concrete until reviewed by Engineer and permission granted to proceed with the repair work;
 - 2) Remove defective concrete down to sound concrete with edges slightly undercut or perpendicular to the surface. Remove a minimum depth of 25mm. No feather edges are permitted; and
 - 3) Prepare surface, prime area and apply specified patching compound in strict accordance with manufacturer's printed instructions.
 - (iii) Structural Concrete Repairs: Work would include areas where both concrete and reinforcement steel has deteriorated:
 - 1) Cut and remove all unsound concrete and reinforcing as directed by the Engineer;
 - 2) Where new sections of reinforcement need to be spliced in, cut back concrete sufficiently to allow proper weld length to existing bars;
 - 3) Where deterioration extends full depth of wall or slab cut key in sound concrete around perimeter of opening;
 - 4) Weld in new sections of reinforcement bars of the same size as existing; and
 - 5) Form opening and fill opening with Type A concrete to which 25 per cent In-Pak grout (by weight) has been added.

G3.20.9 Vertical and Overhead Concrete Repairs

- (a) Concrete surface repair system may either be shotcrete mortar or low pressure spray mortar for structural repairs, and hand or low pressure spray applied polymer modified mortar for non-structural surface applications.
- (b) The Contractor shall carry out the following preparation for repairs:
 - (i) Identify unsound and deteriorated concrete by sounding techniques.

- (ii) Remove unsound, honeycombed, deteriorated, or otherwise defective concrete from work areas by high pressure water blasting machine capable of removing concrete surfaces to a minimum amplitude roughness of 5 millimetres, or as shown on the Drawings. Where final surface is required to be flush with existing surface remove existing concrete to a depth as required for application of minimum thickness of mortar.
- (iii) Do not use power-driven jackhammers and chipping hammers, unless water blasting is not practical or may cause other damage to installed or existing equipment.
- (iv) To avoid tapered shoulders square edges of patch areas by sawing or chipping approximately 100 millimetres beyond area of unsound or deteriorated concrete.
- (v) Collect and dispose of water from removal operations in manner and location acceptable to the Board.
- (vi) Where reinforcing steel is corroded, remove concrete minimum of 25 millimetres clear around reinforcing bar for application and bonding of new mortar to entire surface of exposed reinforcing steel if the following surface conditions exist:
 - 1) 50 per cent or more of reinforcing bar periphery surface is exposed during removal of concrete;
 - 2) 25 per cent or more of reinforcing steel periphery surface is exposed during removal of concrete and extent of reinforcing bar corrosion is such that more than 25 per cent of material is lost; and
 - 3) Visually evident that bond between existing concrete and reinforcement has deteriorated.
- (vii) Clean exposed reinforcing steel of rust and concrete in accordance with recommendations of repair material manufacturer and in accordance with recommendations ICRI 03730.
- (viii) Replace deteriorated reinforcing steel with new reinforcing steel equivalent in cross-sectional area to original reinforcing steel.
- (ix) Coat exposed reinforcing steel with flexible cementitious reinforcing bar coating.
- (x) Keep areas from which concrete has been removed free of dirt, dust, and water blasting slurry.
- (xi) Remove contaminates from prepared areas before new mortar is applied.
- (xii) Clean surface of laitance and contamination prior to applying mortar.
- (xiii) Dampen area to be patched and an area at least 150 millimetres beyond for at least 24 hours to prevent absorption of water from patching mortar to provide saturated surface dry (SSD) condition for existing concrete at time of application of mortar, in accordance with manufacturer's printed instructions.

- (c) Mesh Reinforcement Installation
 - (i) Provide mesh reinforcement when existing reinforcement is not exposed, and when mortar application is more than 75 millimetres deep, unless otherwise shown on the Drawings;
 - (ii) Install mesh anchors in accordance with mesh manufacturer's instructions;
 - (iii) Fasten reinforcement to mesh anchors with tie wire; and
 - (iv) Lap reinforcement a minimum of one mesh spacing and secure with tie wire at intervals no less than 300 millimetres.
- (d) Low Pressure Spray Mortar Application
 - (i) Mix mortar in mortar-concrete mixer attached to pump-spray equipment for spray application. Mix with a slow speed drill and jiffler type paddle or small mortar type mixer for hand trowel application;
 - (ii) After priming substrate per manufacturer's recommendations, apply mortar by low pressure spraying equipment, unless noted otherwise;
 - (iii) Bonding Agent:
 - 1) Use bonding agent for hand applied areas;
 - 2) Application of repair mortar over the bonding agent shall be completed within time frame recommended by bonding agent manufacturer; and
 - 3) Consult with manufacturer for optimum and minimum acceptable degrees of surface tackiness of coat.
 - (iv) Work mortar firmly and quickly into repair area. Finish to match adjacent concrete surface; and
 - (v) Finish exposed mortar surface with steel trowel finish when finish coat is not applied to smooth even surface matching adjacent concrete.
- (e) Polymer-Modified Repair Mortar Application (Non-Structural)
 - (i) Mix mortar in mortar-concrete mixer attached to pump-spray equipment for spray application. Mix with a slow speed drill and jiffler type paddle or small mortar type mixer for hand trowel application;
 - (ii) Hand Trowelling: Apply bond coat of the repair mortar to the SSD prepared substrate before application of mortar. Do not apply more of the bond coat than can be covered with mortar before the bond coat dries. Do not retemper this bond coat;
 - (iii) Place mortar by hand or low pressure spray and trowel to specified surface finish; and
 - (iv) Finish exposed mortar surface with steel trowel finish when finish coat is not applied to smooth even surface matching adjacent concrete.
- (f) Curing
 - (i) Prior to water curing, water fog spray the applied mortar material in accordance with mortar system manufacturer's instructions;
 - (ii) Commence water curing after mortar system application and when curing will not cause erosion of mortar;

- (iii) Continuously cure mortar system for a period of seven days;
- (iv) Do not membrane cure, unless method is part of mortar system manufacturer's instructions and approval is obtained from the Engineer; and
- (v) Cure intermediate layers of mortar in accordance with manufacturer's instructions.

G3.20.10 Concrete Repair and Crack Injection

- (a) References: The following is a list of standards which may be referenced in this Section:
 - (i) BS EN 1504, Repair of concrete structures - Products and systems for the protection and repair of concrete structures;
 - (ii) American Society for Testing and Materials (ASTM);
 - (iii) D638, Standard Test Method for Tensile Properties of Plastics;
 - (iv) D648, Standard Test Method for Deflection of Plastics Under Flexural Load;
 - (v) D695, Standard Test Method for Compressive Properties of Rigid Plastics;
 - (vi) D790, Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials;
 - (vii) American Association of State Highway and Transportation Officials (AASHTO): T237, Standard Method for Testing Epoxy Resin Adhesive; and
 - (viii) American National Standards International/National Sanitation Foundation (ANSI/NSF):
 - 1) Standard 60, Standard for Drinking Water Treatment and Chemicals – Health Effects; and
 - 2) Standard 61, Standard for Drinking Water System Components – Health Effects.
- (b) Definitions
 - (i) Large Cracks: Wider than 0.4mm; and
 - (ii) Small Cracks: Width equal to 0.4mm or less.
- (c) Submittals
 - (i) Shop Drawings:
 - 1) Physical and chemical properties for polyurethane resins; and
 - 2) Technical data for metering, mixing, and injection equipment.
 - (ii) Quality Control Submittals:
 - 1) Manufacturer's recommended surface preparation procedures and application instructions for polyurethane resins;
 - 2) Installation instructions for repairing core holes with epoxy grout;
 - 3) Certificates from Manufacturer's that all products used are NSF 61 approved and is suitable for potable water structure applications;

- 4) Statements of Qualification for polyurethane resins; and
 - 5) Contract Closeout Submittals: polyurethane resins injection pressure test records for concrete crack repair work.
- (d) Quality Assurance
- (i) Qualifications for Injection Staff:
 - 1) Manufacturer's Site Representative:
 - Capable of instructing successful methods for restoring concrete structures utilizing polyurethane injection process;
 - Understands and is capable of explaining technical aspects of correct material selection and use; and
 - Experienced in the operation, maintenance, and troubleshooting of application equipment.
 - 2) Injection crew and job foreman shall provide written and verifiable evidence showing compliance with the following requirements:
 - Licensed and certified by polyurethane manufacturer.
- (e) Performance Requirements
- (i) The injected polyurethane injection resins shall fill the cracks and joints, and in no case shall the depth of penetration of the injection material be less than ninety per cent (90 per cent) of:
 - 1) The full thickness of the concrete section for cracks; and
 - 2) The depth between the waterstop and the inside face of the structure for joints.
 - (ii) The cured injected polyurethane injection resin shall form a dense rubber like flexible foam compression gasket-type seal.
 - (iii) Notwithstanding the foregoing, injected cracks or joints found to exhibit leakage shall be deemed as deficient work irrespective of the depth of penetration observed in quality control core drilled samples. Carry out re-injection of deficient work as required to meet the performance requirements at no cost to the Board.
- (f) Delivery, Storage, And Handling
- (i) Packing and Shipping: Package polyurethane resins material in new sealed containers and label with following information:
 - 1) Manufacturer's name;
 - 2) Product name and lot number; and
 - 3) Mix ratio by volume instructions.
 - (ii) Storage and Protection: Store polyurethane resins materials containers in accordance with manufacturer's printed instructions or in the absence of such information at ambient temperatures above 0 degrees C and below 49 degrees C.

- (g) Products
 - (i) Polyurethane Resin
 - 1) Single-component Diphenylmethane Diisocyanate (MDI) based, water-activated, hydrophobic type resin; and
 - 2) In cured state forms a dense rubber-like closed cell flexible foam compression gasket seal.
 - (ii) Surface Seal
 - 1) Sufficient strength and adhesion for holding injection fittings firmly in-place to resist pressures preventing leakage during injection; and
 - 2) Capable of removal after injection adhesive has cured.
 - (iii) Acid Flushing Solution
 - 1) Pre-mixed solution of food grade phosphoric acid diluted to a 5 per cent plus or minus 0.5 per cent solution, by volume, with potable water prior to delivery to site.
 - (iv) Sample Bottle
 - 1) For polyurethane resin, supply each sample bottle containing a volume of water equal to 5 per cent of the volume of the bottle; and
 - 2) Source Quality Control.
 - (v) Test Requirements: Perform tests for each batch of polyurethane resins.
- (h) Execution
 - (i) General
 - 1) Cracks:
 - Repair by injection of polyurethane resins; and
 - Repair cracks specified or as shown.
 - 2) The width of the cracks shown may vary along the length and through the thickness of the concrete section; and
 - 3) Remove all excess unused polyurethane resins materials inside the structure prior to the end of the work.
 - (ii) Preparation
 - 1) Free cracks from loose matter, dirt, laitance, oil, grease, salt, and other contaminants;
 - 2) Clean cracks in accordance with polyurethane resins manufacturer's printed instructions;
 - 3) Clean surfaces adjacent to cracks from dirt, dust, grease, oil, efflorescence, encrustation, and other foreign matter detrimental to bond of surface seal system in accordance with manufacturer's printed instructions;
 - 4) Do not use acids and corrosives for cleaning, other than those specified herein unless neutralized prior to injecting; and

- 5) Inject cracks with water first before injecting polyurethane grout. Coordinate timing for water to be injected first before injecting polyurethane grout so that required moisture content criteria is satisfied. Confirm this requirement with product manufacturer's representative in writing before injecting any cracks with polyurethane.

(iii) Application

1) Drilling Holes for Injection:

- Drill holes shall be installed along cracks and joints designated for repair as required to meet the performance requirements for injection;
- The requirements for installing drill holes for injection provided below represent acceptable minimum standards of practice;
- Drill holes for injection at an angle between 45 degrees and 30 degrees from perpendicular to the surface of the concrete and perpendicular to the alignment of the or joints, to intersect the cracks at the midpoint of the concrete section, except as noted otherwise;
- Locate the drill holes on alternate sides of the crack or joint where possible, unless the orientation of the crack or joint is known or has been verified by non-destructive testing techniques or core drilling;
- The spacing of the drill holes not to exceed 300 millimetres, except as noted otherwise;
- Adjust the location and angle of the drill holes to suit the orientation of the crack or joint and at the required angle, such as using a template, during the work especially at the commencement of drilling holes for injection and at the beginning of each subsequent shift;
- Take measures to prevent drilling the holes for injection too shallow, or too deep, and/or damaging the existing waterstop in the joints;
- Remove dust and debris in the drill holes and on the interior surface of the structure resulting from the drilling operation, by flushing with water prior to installing the injection packers or ports;
- Install injection packers or ports in the drill holes in accordance with the manufacturer's printed instructions with the zerk coupling, or other one way ball or check valve, to permit testing for watertightness and acid flushing of the cracks and joints; and
- Inject cracks with water first before injecting polyurethane grout. Coordinate timing for the water to be injected first before injecting polyurethane grout so that required moisture content criteria is satisfied in crack and the crack is ready to inject polyurethane. Confirm above requirement with product manufacturer's representative in writing before injecting any cracks with polyurethane.

- 2) Application of Surface Seal along Cracks and Joints:
 - Apply a surface seal along the length of the designated cracks and joints on the top side and underside of slabs, or both sides of walls, as required in order to contain and prevent escape of the polyurethane injection resins in accordance with manufacturer's printed instructions; and
 - Cure the surface seal in accordance with manufacturer's printed instructions.
- 3) Mixing Polyurethane Injection Resin:
 - Add accelerator to the polyurethane injection resin at the required dosage to produce a cured polyurethane material meeting the performance requirements, and mix thoroughly in accordance with the manufacturer's printed instructions until a homogeneous mixture is obtained outside of the structure; and
 - Heat the polyurethane injection resin materials prior to and during the mixing and injection as required and in accordance with manufacturer's printed instructions to a temperature between 30 degrees C and 55 degrees C. Injection of polyurethane injection resin when it is less than 30 degrees C or more than 55 degrees C is prohibited.
- 4) Injection of Cracks and Joints:
 - Inject the polyurethane injection resins with water, or in a neat form into cracks and joints in a sequential manner, and re-inject as required, to meet the performance requirements;
 - Start injection into each crack or joint at lowest elevation entry port for vertical cracks and joints and at one end for horizontal cracks and joints in vertical structure. Continue injection at first port until resin begins to flow out of next port. Plug first port and start injection at second port until polyurethane resin flows from next port. Inject crack or joint in a sequential manner until the full length of the crack or joint has been injected. At no time inject more than 1800mm length of vertical crack or joint before verifying that the first 300mm of injected crack or joint is set and cured; and
 - Prior to commencing the injection work along a crack or joint in horizontal structure when processed with zerk couplings are used, remove the zerk couplings from the injection packers or ports except for the two packers located where the injection work will commence. Commence injection work in the first two packers. Once clean resin is vented from the third injection packer, cease injection at the first packer, and install the zerk coupling and commence injection at the third packer. Repeat the process for the fourth and subsequent packers until the full length of the crack or joint has been injected.

5) Finishing:

- Cure polyurethane resins after crack or joints have been completely filled to allow surface seal removal without draining or runback of polyurethane resin material from crack or joints;
- Remove surface seal and injection packers or ports from along the crack or joints;
- Finish face flush with adjacent concrete;
- Indentations or protrusions caused by placement of entry ports drill holes, or damage from removal of surface seal are not acceptable;
- Grind off protrusions, and patch indentations and holes from packers and ports with a suitable patch material to the satisfaction of the Engineer; and
- Remove surplus surface seal material on splatters and injection material runs and spills from concrete surfaces.

(iv) Equipment

- 1) Portable, positive displacement type pumps with in-line metering to meter and mix two adhesive components, and inject mixture into crack or joint.
- 2) Polyurethane Injection Equipment:
 - Positive displacement type pump;
 - Equip the polyurethane injection equipment with pail heater(s) suitable for plastic pails capable of maintaining the polyurethane resin and accelerator mixture between 10 and 70 degrees C; and
 - A thermometer with each pump for monitoring the temperature of the polyurethane resin.
- 3) Discharge Pressure: Automatic pressure controls capable of discharging injection material at pressures up to 1400 kPa, plus or minus 5 per cent, and able to maintain pressure.

(v) Field Quality Control

- 1) Injection Pressure Test:
 - Disconnect mixing head of injection equipment and connect two adhesives component delivery lines to pressure check device.
- 2) Pressure Check Device:
 - Two independent valved nozzles capable of controlling flow rate and pressure by opening or closing of valve.
- 3) Pressure gauge capable of sensing pressure build-up behind each valve.
- 4) Close valves on pressure check device and operate equipment until gauge pressure on each line reads 1100 kPa.
- 5) Stop pumps and observe pressure; do not allow pressure gauge to drop below 1050 kPa within three minutes.

- 6) Run pressure test for each injection equipment unit:
 - Beginning and end of each injection work day; and
 - When injection work as stop for more than 45 minutes.
- 7) Check tolerance to verify equipment capable of meeting specified ratio tolerance.
- 8) Bottled Samples:
 - During the injection operation, provide at least one sample of the mixed polyurethane injection resins per shift per injection pump in a sample bottle;
 - For polyurethane injection, half fill each sample bottle, containing a small amount of water (5 per cent by volume) with polyurethane injection resin material from the pump discharge hose. Swirl or lightly shake sample to thoroughly mix resin with the water. When foaming rises near the top of the bottle, install the cap to contain the expanding resin material;
 - Place the filled sample bottle upright in a box or pail and allow the sample to cure. After sample has been allowed to cure, cut the bottled sample open and visually inspect the contents to verify that the polyurethane injection resins material has completely reacted and cured;
 - Label each bottled sample with the Contractor's name, date and time sample was taken, and the location in the structure where the sample was taken. Record details of the bottle samples;
 - Should a bottled sample(s) indicate a problem with the polyurethane injection resins, take verifying core sample immediately from the cracks, where the material was used;
 - Should the above-referenced bottle sample(s) and core sample(s) indicate a problem with the polyurethane injection resins, arrange to have a Technical Representative of the polyurethane injection resins manufacturer come to site to review the bottled sample(s) and core drilled sample(s) with the Engineer and provide technical advice on corrective measures; and
 - Carry out further investigation work or corrective measures recommended by the Technical Representative of the polyurethane injection resins manufacturer.

G3.20.11 Waterproof Concrete

- (a) The performance of waterproof concrete shall meet the following requirements:
 - (i) Absorption (BS 1181, Part 122) of <1%;
 - (ii) Water Penetration, (Din 1048, or BS EN 12390-8) of <10mm; and
 - (iii) Permeability, (ASTM C1202) of <1000 coulombs

G3.20.12 Concrete Exposed to Sewer

- (a) All concrete used for structural elements should be sulphate-resisting concrete using sulphate-resisting cement or equivalent.
- (b) Concrete for water retaining structure and expose to corrosive agent or in aggressive environment, i.e. containing sewer or exposed to the effects of H₂S shall have silica fume, conforming to BS EN 13263, of minimum 7 percent of cement content by weight and high range superplasticiser. The performance of such concrete shall meet the G3.20.11 Waterproof Concrete requirements.

G3.20.13 Water Testing of Liquid Retaining Structures

- (a) All compartments of the liquid retaining structures shall be water tested by the Contractor prior to commissioning and use. No such structure shall be commissioned or used until the Contractor has demonstrated that all compartments of that structure are watertight.
- (b) Water testing of each liquid retaining structure shall commence as soon as practicable after completion of that structure, but only after the Contractor has demonstrated to the satisfaction of the S.O. that the structure has attained its full design concrete strength. Water testing shall not occur without the written consent of the S.O..
- (c) Each compartment of a liquid retaining structure shall be tested separately, and adjacent compartments shall not be tested concurrently.
- (d) The water test shall comprise the following:
 - (i) Each compartment shall be filled gradually with water to the maximum operating water level indicated on the hydraulic profile. This shall occur at a steady rate not exceeding 1m increase in water depth per 12 hours;
 - (ii) Following the first fill, the structure shall remain filled with water for a standing period of seven days to allow the concrete to absorb any water. The maximum operating water level shall be maintained with the addition of water as necessary; and
 - (iii) Following the above standing period, the compartment shall remain filled with water for a further test period of seven days. During this test, the drop in water level during the test period, allowing for evaporation and rainfall ingress, shall not exceed 0.2% of the maximum operating water depth for that compartment.
- (g) Each compartment shall only be deemed to have passed the water test once the S.O. provides written confirmation that he is satisfied that it complies with the above requirements. Each liquid retaining structure shall only be deemed to have passed the water test once all its compartments have passed the water test.
- (h) If a compartment does not pass the water test, the Contractor shall promptly determine the source of leakage, inspecting all exposed surfaces for damp patches. The source may also be determined by reducing the water level progressively over a time period.

- (i) Following determination of the source(s) of leakage, the Contractor shall propose appropriate remedial work for approval by the S.O. No such remedial work shall be carried out without the approval of the S.O.
- (j) Following completion of the remedial work, the water test shall be repeated. If necessary, further remedial work and water testing shall be carried out until such time as each compartment passes the water test.
- (k) Water used for water testing shall be re-used for subsequent water testing as many times as reasonably practicable.

END OF SECTION